

SAT Sharks - Demo Full Test - Math

3 modules x 22 questions (6 grid-in per module). Module 1 for everyone; Module 2 Easy if below 65% on Module 1, Module 2 Hard if 65% or above.

DEMO CONTENT - sample questions written to show the SAT Sharks full-test upload format.

SECTION: MATH

MODULE 1

QUESTION 1

CATEGORY: SAT Algebra

DIFFICULTY: MEDIUM

PROMPT: If $8x + 14 = 86$, what is the value of x ?

A: 11

B: 23

C: 72

D: 9

ANSWER: D

EXPLANATION: Subtract 14 from both sides to get $8x = 72$, then divide by 8: $x = 9$.

END QUESTION

QUESTION 2

CATEGORY: SAT Data & Statistics

DIFFICULTY: MEDIUM

PROMPT: What is 60% of 40?

A: 240

B: 64

C: 16

D: 24

ANSWER: D

EXPLANATION: $60\% \text{ of } 40 = 0.6 \times 40 = 24$.

END QUESTION

QUESTION 3**CATEGORY:** SAT Geometry**DIFFICULTY:** MEDIUM

PROMPT: A rectangle has a length of 11 centimeters and a width of 7 centimeters. What is the area of the rectangle, in square centimeters?

A: 77**B:** 18**C:** 88**D:** 36**ANSWER:** A**EXPLANATION:** Area = length $\times$ width = $11 \times 7 = 77$.**END QUESTION****QUESTION 4****CATEGORY:** SAT Algebra**DIFFICULTY:** MEDIUM

PROMPT: A gym charges a one-time fee of \$90 plus \$24 per month. Which expression gives the total cost, in dollars, of a membership for n months?

A: $24n + 90$ **B:** $90n + 24$ **C:** $114n$ **D:** $24(n + 90)$ **ANSWER:** A**EXPLANATION:** The monthly charge is $24n$ and the one-time fee 90 is added once: $24n + 90$.**END QUESTION****QUESTION 5****CATEGORY:** SAT Advanced Math**DIFFICULTY:** MEDIUM

PROMPT: The function f is defined by $f(x) = 3x^2 - 3x + 6$. What is the value of $f(2)$?

A: 13**B:** 24**C:** 12**D:** 11**ANSWER:** C**EXPLANATION:** $f(2) = 3(2)^2 - 3(2) + 6 = 12 - 6 + 6 = 12$.**END QUESTION**

QUESTION 6**CATEGORY:** SAT Advanced Math**DIFFICULTY:** MEDIUM**TYPE:** GRID_IN**PROMPT:** If $x^7 \cdot x^8 = x^n$ for all positive values of x , what is the value of n ?**ANSWER:** 15**EXPLANATION:** When multiplying powers with the same base, add the exponents: $7 + 8 = 15$.**END QUESTION****QUESTION 7****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** MEDIUM**PROMPT:** A bag contains 9 red, 3 blue, and 2 green marbles. If one marble is chosen at random, what is the probability that it is blue?**A:** $3/11$ **B:** $3/14$ **C:** $1/3$ **D:** $9/14$ **ANSWER:** B**EXPLANATION:** There are 3 blue marbles out of 14 in all, so the probability is $3/14 = 3/14$.**END QUESTION****QUESTION 8****CATEGORY:** SAT Algebra**DIFFICULTY:** MEDIUM**PROMPT:** What is the slope of the line that passes through the points $(0, 4)$ and $(3, 10)$?**A:** 6**B:** 3**C:** 2**D:** -2**ANSWER:** C**EXPLANATION:** Slope = $(10 - 4) / (3 - 0) = 6 / 3 = 2$.**END QUESTION****QUESTION 9****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** MEDIUM**TYPE:** GRID_IN**PROMPT:** What is the mean of the data set 12, 12, 6, 7, 8?**ANSWER:** 9**EXPLANATION:** The sum is 45 and there are 5 values, so the mean is $45 / 5 = 9$.**END QUESTION**

QUESTION 10**CATEGORY:** SAT Geometry**DIFFICULTY:** MEDIUM**PROMPT:** A circle has a radius of 5. What is the area of the circle?**A:** 100π **B:** 5π **C:** 10π **D:** 25π **ANSWER:** D**EXPLANATION:** $\text{Area} = \pi r^2 = \pi(5)^2 = 25\pi$.**END QUESTION****QUESTION 11****CATEGORY:** SAT Algebra**DIFFICULTY:** MEDIUM**PROMPT:** If $x + y = 13$ and $x - y = -1$, what is the value of x ?**A:** 13**B:** 7**C:** 6**D:** -1**ANSWER:** C**EXPLANATION:** Add the equations: $2x = 12$, so $x = 6$.**END QUESTION****QUESTION 12****CATEGORY:** SAT Geometry**DIFFICULTY:** MEDIUM**PROMPT:** A right triangle has legs of length 3 and 4. What is the length of the hypotenuse?**A:** 5**B:** 4**C:** 6**D:** 7**ANSWER:** A**EXPLANATION:** By the Pythagorean theorem, $c^2 = 3^2 + 4^2 = 25$, so $c = 5$.**END QUESTION**

QUESTION 13**CATEGORY:** SAT Advanced Math**DIFFICULTY:** MEDIUM**PROMPT:** What is the sum of the solutions to the equation $x^2 - 3x + 2 = 0$?**A:** 2**B:** 1**C:** 3**D:** -3**ANSWER:** C**EXPLANATION:** Factor: $(x - 1)(x - 2) = 0$, so $x = 1$ or $x = 2$. Their sum is 3.**END QUESTION****QUESTION 14****CATEGORY:** SAT Geometry**DIFFICULTY:** MEDIUM**TYPE:** GRID_IN**PROMPT:** In a triangle, two of the angles measure 40 degrees and 69 degrees. What is the measure, in degrees, of the third angle?**ANSWER:** 71**EXPLANATION:** The angles of a triangle add to 180 degrees: $180 - 40 - 69 = 71$.**END QUESTION****QUESTION 15****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** MEDIUM**PROMPT:** What is the median of the data set 3, 10, 4, 1, 34, 2, 12?**A:** 4**B:** 3**C:** 9**D:** 1**ANSWER:** A**EXPLANATION:** In order: 1, 2, 3, 4, 10, 12, 34. The middle (4th) value is 4.**END QUESTION**

QUESTION 16**CATEGORY:** SAT Advanced Math**DIFFICULTY:** MEDIUM**PROMPT:** A population of 300 bacteria doubles every hour. Which expression gives the number of bacteria after t hours?**A:** $600t$ **B:** $300(2)^t$ **C:** $300(t)^2$ **D:** $300 + 2t$ **ANSWER:** B**EXPLANATION:** Doubling each hour multiplies the population by 2 once per hour, so after t hours it is $300(2)^t$.**END QUESTION****QUESTION 17****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** MEDIUM**TYPE:** GRID_IN**PROMPT:** The price of a jacket was \$80. The price was increased by 20%, and then the new price was decreased by 25%. What is the final price of the jacket, in dollars?**ANSWER:** 72**EXPLANATION:** After the increase: $80 \times 1.2 = 96$. After the decrease: $96 \times 0.75 = 72$.**END QUESTION****QUESTION 18****CATEGORY:** SAT Algebra**DIFFICULTY:** MEDIUM**PROMPT:** If $8x + 5 = 85$, what is the value of x ?**A:** 11**B:** 80**C:** 15**D:** 10**ANSWER:** D**EXPLANATION:** Subtract 5 from both sides to get $8x = 80$, then divide by 8: $x = 10$.**END QUESTION****QUESTION 19****CATEGORY:** SAT Geometry**DIFFICULTY:** MEDIUM**TYPE:** GRID_IN**PROMPT:** A rectangular box has a length of 9 inches, a width of 8 inches, and a height of 4 inches. What is the volume of the box, in cubic inches?**ANSWER:** 288**EXPLANATION:** Volume = length $\times$ width $\times$ height = $9 \times 8 \times 4 = 288$.**END QUESTION**

QUESTION 20**CATEGORY:** SAT Advanced Math**DIFFICULTY:** MEDIUM**PROMPT:** The function f is defined by $f(x) = (x - 1)^2 + 5$. What is the minimum value of $f(x)$?**A:** 1**B:** 6**C:** -5**D:** 5**ANSWER:** D**EXPLANATION:** A squared term is never negative, so $f(x)$ is smallest when $x = 1$. Then $f(1) = 0 + 5 = 5$.**END QUESTION****QUESTION 21****CATEGORY:** SAT Algebra**DIFFICULTY:** MEDIUM**TYPE:** GRID_IN**PROMPT:** If $x + y = 11$ and $x - y = -7$, what is the value of x ?**ANSWER:** 2**EXPLANATION:** Add the equations: $2x = 4$, so $x = 2$.**END QUESTION****QUESTION 22****CATEGORY:** SAT Geometry**DIFFICULTY:** MEDIUM**PROMPT:** A right circular cylinder has a radius of 6 and a height of 9. What is the volume of the cylinder?**A:** 54π **B:** 648π **C:** 324π **D:** 108π **ANSWER:** C**EXPLANATION:** Volume = $\pi r^2 h = \pi(6)^2(9) = 324\pi$.**END QUESTION****END MODULE**

MODULE 2 EASY

QUESTION 1

CATEGORY: SAT Algebra

DIFFICULTY: EASY

PROMPT: If $3x + 7 = 28$, what is the value of x ?

A: 21

B: 7

C: 14

D: 9

ANSWER: B

EXPLANATION: Subtract 7 from both sides to get $3x = 21$, then divide by 3: $x = 7$.

END QUESTION

QUESTION 2

CATEGORY: SAT Data & Statistics

DIFFICULTY: EASY

PROMPT: What is 25% of 200?

A: 500

B: 45

C: 50

D: 150

ANSWER: C

EXPLANATION: $25\% \text{ of } 200 = 0.25 \times 200 = 50$.

END QUESTION

QUESTION 3

CATEGORY: SAT Geometry

DIFFICULTY: EASY

PROMPT: A rectangle has a length of 9 centimeters and a width of 4 centimeters. What is the area of the rectangle, in square centimeters?

A: 45

B: 26

C: 36

D: 13

ANSWER: C

EXPLANATION: $\text{Area} = \text{length} \times \text{width} = 9 \times 4 = 36$.

END QUESTION

QUESTION 4**CATEGORY:** SAT Algebra**DIFFICULTY:** EASY**PROMPT:** A gym charges a one-time fee of \$60 plus \$32 per month. Which expression gives the total cost, in dollars, of a membership for n months?**A:** $60n + 32$ **B:** $32(n + 60)$ **C:** $92n$ **D:** $32n + 60$ **ANSWER:** D**EXPLANATION:** The monthly charge is $32n$ and the one-time fee 60 is added once: $32n + 60$.**END QUESTION****QUESTION 5****CATEGORY:** SAT Advanced Math**DIFFICULTY:** EASY**PROMPT:** The function f is defined by $f(x) = x^2 - 6x + 6$. What is the value of $f(4)$?**A:** 46**B:** -10**C:** -2**D:** -1**ANSWER:** C**EXPLANATION:** $f(4) = (4)^2 - 6(4) + 6 = 16 - 24 + 6 = -2$.**END QUESTION****QUESTION 6****CATEGORY:** SAT Advanced Math**DIFFICULTY:** EASY**TYPE:** GRID_IN**PROMPT:** If $x^7 * x^3 = x^n$ for all positive values of x , what is the value of n ?**ANSWER:** 10**EXPLANATION:** When multiplying powers with the same base, add the exponents: $7 + 3 = 10$.**END QUESTION**

QUESTION 7**CATEGORY:** SAT Data & Statistics**DIFFICULTY:** EASY**PROMPT:** A bag contains 8 red, 6 blue, and 6 green marbles. If one marble is chosen at random, what is the probability that it is blue?**A:** $6/14$ **B:** $1/6$ **C:** $2/5$ **D:** $3/10$ **ANSWER:** D**EXPLANATION:** There are 6 blue marbles out of 20 in all, so the probability is $6/20 = 3/10$.**END QUESTION****QUESTION 8****CATEGORY:** SAT Algebra**DIFFICULTY:** EASY**PROMPT:** What is the slope of the line that passes through the points (3, -5) and (5, 3)?**A:** -4**B:** 4**C:** 8**D:** 2**ANSWER:** B**EXPLANATION:** Slope = $(3 - -5) / (5 - 3) = 8 / 2 = 4$.**END QUESTION****QUESTION 9****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** EASY**TYPE:** GRID_IN**PROMPT:** What is the mean of the data set 13, 14, 28, 30, 15?**ANSWER:** 20**EXPLANATION:** The sum is 100 and there are 5 values, so the mean is $100 / 5 = 20$.**END QUESTION**

QUESTION 10**CATEGORY:** SAT Geometry**DIFFICULTY:** EASY**PROMPT:** A circle has a radius of 3. What is the area of the circle?**A:** 36π **B:** 9π **C:** 6π **D:** 3π **ANSWER:** B**EXPLANATION:** $\text{Area} = \pi r^2 = \pi(3)^2 = 9\pi$.**END QUESTION****QUESTION 11****CATEGORY:** SAT Algebra**DIFFICULTY:** EASY**PROMPT:** If $x + y = 7$ and $x - y = -1$, what is the value of x ?**A:** 7**B:** 4**C:** 3**D:** -1**ANSWER:** C**EXPLANATION:** Add the equations: $2x = 6$, so $x = 3$.**END QUESTION****QUESTION 12****CATEGORY:** SAT Geometry**DIFFICULTY:** EASY**PROMPT:** A right triangle has legs of length 8 and 15. What is the length of the hypotenuse?**A:** 17**B:** 23**C:** 16**D:** 18**ANSWER:** A**EXPLANATION:** By the Pythagorean theorem, $c^2 = 8^2 + 15^2 = 289$, so $c = 17$.**END QUESTION**

QUESTION 13**CATEGORY:** SAT Advanced Math**DIFFICULTY:** EASY**PROMPT:** What is the sum of the solutions to the equation $x^2 - 10x + 21 = 0$?**A:** -10**B:** 21**C:** 10**D:** 4**ANSWER:** C**EXPLANATION:** Factor: $(x - 3)(x - 7) = 0$, so $x = 3$ or $x = 7$. Their sum is 10.**END QUESTION****QUESTION 14****CATEGORY:** SAT Geometry**DIFFICULTY:** EASY**TYPE:** GRID_IN**PROMPT:** In a triangle, two of the angles measure 63 degrees and 63 degrees. What is the measure, in degrees, of the third angle?**ANSWER:** 54**EXPLANATION:** The angles of a triangle add to 180 degrees: $180 - 63 - 63 = 54$.**END QUESTION****QUESTION 15****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** EASY**PROMPT:** What is the median of the data set 38, 13, 38, 1, 22, 35, 33?**A:** 1**B:** 26**C:** 33**D:** 22**ANSWER:** C**EXPLANATION:** In order: 1, 13, 22, 33, 35, 38, 38. The middle (4th) value is 33.**END QUESTION**

QUESTION 16**CATEGORY:** SAT Advanced Math**DIFFICULTY:** EASY**PROMPT:** A population of 400 bacteria doubles every hour. Which expression gives the number of bacteria after t hours?**A:** $400 + 2t$ **B:** $400(t)^2$ **C:** $400(2)^t$ **D:** $800t$ **ANSWER:** C**EXPLANATION:** Doubling each hour multiplies the population by 2 once per hour, so after t hours it is $400(2)^t$.**END QUESTION****QUESTION 17****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** EASY**TYPE:** GRID_IN**PROMPT:** The price of a jacket was \$120. The price was increased by 25%, and then the new price was decreased by 20%. What is the final price of the jacket, in dollars?**ANSWER:** 120**EXPLANATION:** After the increase: $120 \times 1.25 = 150$. After the decrease: $150 \times 0.8 = 120$.**END QUESTION****QUESTION 18****CATEGORY:** SAT Algebra**DIFFICULTY:** EASY**PROMPT:** If $4x + 16 = 36$, what is the value of x ?**A:** 9**B:** 21**C:** 20**D:** 5**ANSWER:** D**EXPLANATION:** Subtract 16 from both sides to get $4x = 20$, then divide by 4: $x = 5$.**END QUESTION**

QUESTION 19**CATEGORY:** SAT Geometry**DIFFICULTY:** EASY**TYPE:** GRID_IN

PROMPT: A rectangular box has a length of 9 inches, a width of 6 inches, and a height of 4 inches. What is the volume of the box, in cubic inches?

ANSWER: 216**EXPLANATION:** Volume = length $\times$ width $\times$ height = $9 \times 6 \times 4 = 216$.**END QUESTION****QUESTION 20****CATEGORY:** SAT Advanced Math**DIFFICULTY:** EASY

PROMPT: The function f is defined by $f(x) = (x - 6)^2 + 6$. What is the minimum value of $f(x)$?

A: 12**B:** 6**C:** -6**D:** 7**ANSWER:** B

EXPLANATION: A squared term is never negative, so $f(x)$ is smallest when $x = 6$. Then $f(6) = 0 + 6 = 6$.

END QUESTION**QUESTION 21****CATEGORY:** SAT Algebra**DIFFICULTY:** EASY**TYPE:** GRID_IN

PROMPT: If $x + y = 10$ and $x - y = 4$, what is the value of x ?

ANSWER: 7

EXPLANATION: Add the equations: $2x = 14$, so $x = 7$.

END QUESTION**QUESTION 22****CATEGORY:** SAT Geometry**DIFFICULTY:** EASY

PROMPT: A right circular cylinder has a radius of 5 and a height of 10. What is the volume of the cylinder?

A: 50π **B:** 500π **C:** 250π **D:** 100π **ANSWER:** C

EXPLANATION: Volume = $\pi r^2 h = \pi(5)^2(10) = 250\pi$.

END QUESTION**END MODULE**

MODULE 2 HARD

QUESTION 1

CATEGORY: SAT Algebra

DIFFICULTY: HARD

PROMPT: If $5(x - 3) = 3x - 11$, what is the value of x ?

A: -1

B: 5

C: -2

D: 2

ANSWER: D

EXPLANATION: Distribute to get $5x - 15 = 3x - 11$. Then $2x = 4$, so $x = 2$.

END QUESTION

QUESTION 2

CATEGORY: SAT Data & Statistics

DIFFICULTY: HARD

PROMPT: What is 40% of 140?

A: 54

B: 56

C: 84

D: 560

ANSWER: B

EXPLANATION: 40% of 140 = $0.4 \times 140 = 56$.

END QUESTION

QUESTION 3

CATEGORY: SAT Geometry

DIFFICULTY: HARD

PROMPT: A rectangle has a length of 16 centimeters and a width of 11 centimeters. What is the area of the rectangle, in square centimeters?

A: 192

B: 54

C: 176

D: 27

ANSWER: C

EXPLANATION: Area = length $\times$ width = $16 \times 11 = 176$.

END QUESTION

QUESTION 4**CATEGORY:** SAT Algebra**DIFFICULTY:** HARD**PROMPT:** A gym charges a one-time fee of \$80 plus \$32 per month. Which expression gives the total cost, in dollars, of a membership for n months?**A:** $32n + 80$ **B:** $112n$ **C:** $32(n + 80)$ **D:** $80n + 32$ **ANSWER:** A**EXPLANATION:** The monthly charge is $32n$ and the one-time fee 80 is added once: $32n + 80$.**END QUESTION****QUESTION 5****CATEGORY:** SAT Advanced Math**DIFFICULTY:** HARD**PROMPT:** The function f is defined by $f(x) = x^2 - 4x + 3$. What is the value of $f(4)$?**A:** -5**B:** 4**C:** 3**D:** 35**ANSWER:** C**EXPLANATION:** $f(4) = (4)^2 - 4(4) + 3 = 16 - 16 + 3 = 3$.**END QUESTION****QUESTION 6****CATEGORY:** SAT Advanced Math**DIFFICULTY:** HARD**TYPE:** GRID_IN**PROMPT:** If $(x^4)^3 \cdot x^7 = x^n$ for all positive values of x , what is the value of n ?**ANSWER:** 19**EXPLANATION:** $(x^4)^3 = x^{12}$. Multiplying by x^7 adds exponents: $12 + 7 = 19$.**END QUESTION**

QUESTION 7**CATEGORY:** SAT Data & Statistics**DIFFICULTY:** HARD**PROMPT:** A bag contains 7 red, 9 blue, and 2 green marbles. If one marble is chosen at random, what is the probability that it is blue?**A:** $9/9$ **B:** $1/9$ **C:** $1/2$ **D:** $7/18$ **ANSWER:** C**EXPLANATION:** There are 9 blue marbles out of 18 in all, so the probability is $9/18 = 1/2$.**END QUESTION****QUESTION 8****CATEGORY:** SAT Algebra**DIFFICULTY:** HARD**PROMPT:** What is the slope of the line that passes through the points (2, 6) and (4, -2)?**A:** 2**B:** -4**C:** -8**D:** 4**ANSWER:** B**EXPLANATION:** Slope = $(-2 - 6) / (4 - 2) = -8 / 2 = -4$.**END QUESTION****QUESTION 9****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** HARD**TYPE:** GRID_IN**PROMPT:** What is the mean of the data set 3, 17, 22, 22, 22, 4?**ANSWER:** 15**EXPLANATION:** The sum is 90 and there are 6 values, so the mean is $90 / 6 = 15$.**END QUESTION**

QUESTION 10**CATEGORY:** SAT Geometry**DIFFICULTY:** HARD**PROMPT:** A circle has a radius of 4. What is the area of the circle?**A:** 4π **B:** 64π **C:** 8π **D:** 16π **ANSWER:** D**EXPLANATION:** $\text{Area} = \pi r^2 = \pi(4)^2 = 16\pi$.**END QUESTION****QUESTION 11****CATEGORY:** SAT Algebra**DIFFICULTY:** HARD**PROMPT:** In the system of equations $3x + 5y = 32$ and $x - y = 0$, what is the value of y ?**A:** 5**B:** 4**C:** 3**D:** 0**ANSWER:** B**EXPLANATION:** From $x - y = 0$, $x = y + 0$. Substitute: $3(y + 0) + 5y = 32$, so $8y = 32$ and $y = 4$.**END QUESTION****QUESTION 12****CATEGORY:** SAT Geometry**DIFFICULTY:** HARD**PROMPT:** A right triangle has legs of length 9 and 12. What is the length of the hypotenuse?**A:** 15**B:** 18**C:** 13**D:** 21**ANSWER:** A**EXPLANATION:** By the Pythagorean theorem, $c^2 = 9^2 + 12^2 = 225$, so $c = 15$.**END QUESTION**

QUESTION 13**CATEGORY:** SAT Advanced Math**DIFFICULTY:** HARD**PROMPT:** What is the sum of the solutions to the equation $x^2 - 12x + 11 = 0$?**A:** 12**B:** 10**C:** 11**D:** -12**ANSWER:** A**EXPLANATION:** Factor: $(x - 1)(x - 11) = 0$, so $x = 1$ or $x = 11$. Their sum is 12.**END QUESTION****QUESTION 14****CATEGORY:** SAT Geometry**DIFFICULTY:** HARD**TYPE:** GRID_IN**PROMPT:** In a triangle, two of the angles measure 45 degrees and 65 degrees. What is the measure, in degrees, of the third angle?**ANSWER:** 70**EXPLANATION:** The angles of a triangle add to 180 degrees: $180 - 45 - 65 = 70$.**END QUESTION****QUESTION 15****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** HARD**PROMPT:** What is the median of the data set 29, 8, 32, 3, 36, 40, 4?**A:** 8**B:** 29**C:** 22**D:** 3**ANSWER:** B**EXPLANATION:** In order: 3, 4, 8, 29, 32, 36, 40. The middle (4th) value is 29.**END QUESTION**

QUESTION 16**CATEGORY:** SAT Advanced Math**DIFFICULTY:** HARD**PROMPT:** A population of 800 bacteria doubles every hour. Which expression gives the number of bacteria after t hours?**A:** $800 + 2t$ **B:** $1600t$ **C:** $800(t)^2$ **D:** $800(2)^t$ **ANSWER:** D**EXPLANATION:** Doubling each hour multiplies the population by 2 once per hour, so after t hours it is $800(2)^t$.**END QUESTION****QUESTION 17****CATEGORY:** SAT Data & Statistics**DIFFICULTY:** HARD**TYPE:** GRID_IN**PROMPT:** The price of a jacket was \$120. The price was increased by 25%, and then the new price was decreased by 20%. What is the final price of the jacket, in dollars?**ANSWER:** 120**EXPLANATION:** After the increase: $120 \times 1.25 = 150$. After the decrease: $150 \times 0.8 = 120$.**END QUESTION****QUESTION 18****CATEGORY:** SAT Algebra**DIFFICULTY:** HARD**PROMPT:** If $6(x - 6) = 2x - 12$, what is the value of x ?**A:** 0**B:** 6**C:** -6**D:** 12**ANSWER:** B**EXPLANATION:** Distribute to get $6x - 36 = 2x - 12$. Then $4x = 24$, so $x = 6$.**END QUESTION**

QUESTION 19**CATEGORY:** SAT Geometry**DIFFICULTY:** HARD**TYPE:** GRID_IN

PROMPT: A rectangular box has a length of 8 inches, a width of 4 inches, and a height of 4 inches. What is the volume of the box, in cubic inches?

ANSWER: 128**EXPLANATION:** Volume = length $\times$ width $\times$ height = $8 \times 4 \times 4 = 128$.**END QUESTION****QUESTION 20****CATEGORY:** SAT Advanced Math**DIFFICULTY:** HARD

PROMPT: The function f is defined by $f(x) = (x - 7)^2 + 2$. What is the minimum value of $f(x)$?

A: 2**B:** 9**C:** -2**D:** 7**ANSWER:** A

EXPLANATION: A squared term is never negative, so $f(x)$ is smallest when $x = 7$. Then $f(7) = 0 + 2 = 2$.

END QUESTION**QUESTION 21****CATEGORY:** SAT Algebra**DIFFICULTY:** HARD**TYPE:** GRID_IN

PROMPT: In the system of equations $4x + 2y = 38$ and $x - y = -4$, what is the value of y ?

ANSWER: 9

EXPLANATION: From $x - y = -4$, $x = y + -4$. Substitute: $4(y + -4) + 2y = 38$, so $6y = 54$ and $y = 9$.

END QUESTION

QUESTION 22

CATEGORY: SAT Geometry

DIFFICULTY: HARD

PROMPT: A right circular cylinder has a radius of 5 and a height of 6. What is the volume of the cylinder?

A: 300π

B: 60π

C: 30π

D: 150π

ANSWER: D

EXPLANATION: Volume = $\pi r^2 h = \pi(5)^2(6) = 150\pi$.

END QUESTION

END MODULE